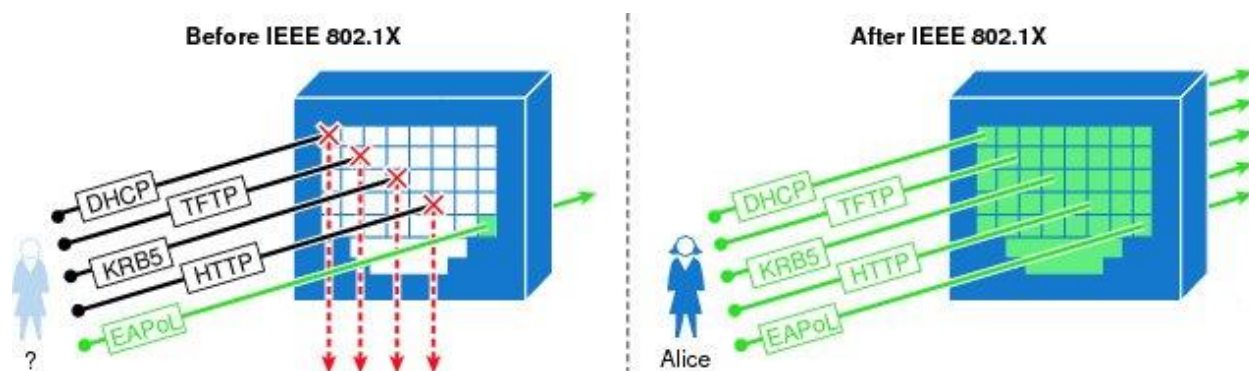


Implement 802.1X:

- o IEEE 802.1x is a standard set by the IEEE 802.1 working group.
- o IEEE 802.1X commonly referred or called Dot1x.
- o IEEE 802.1X authentication is a Data Link Layer (Layer 2) protocol.
- o IEEE 802.1X provide port-based network access control using authentication.
- o IEEE 802.1X authentication method service is called port-level authentication.
- o IEEE 802.1x is defined as a standard for “Port-Based Network Access Control”.
- o The protocol in 802.1X is called EAP Encapsulation over LANs (EAPOL).
- o IEEE 802.1X method is used mainly for port-based authentication.
- o Dot1x can be used to prevent unauthorized devices from gaining access to network.
- o Dot1x standards provide authentication & authorization services at network port level.
- o Dot1x provide real security for wired and wireless networks at layer two.
- o IEEE 802.1x authentication is a Client and Server based authentication protocol.



802.1X Components:

The IEEE 802.1x framework defines three roles in the authentication process:

Supplicant (Client or Host):

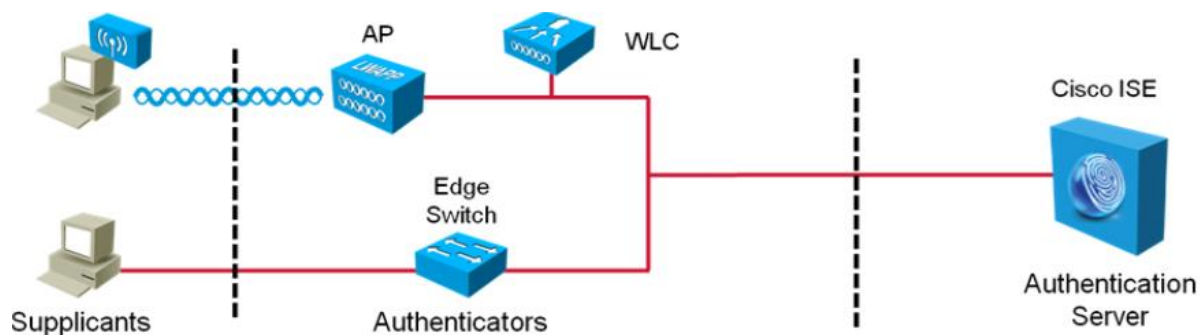
- o Supplicant is the user or device that wants access the wireless or wire network.
- o Supplicant, client or host is the device or user requiring authentication.
- o Supplicant is also known Client, as 802.1x Port-Based Authentication.
- o Supplicant is the Workstation that is connected through Network Access Switch.
- o Supplicant is the workstation request for accessing the network resources.
- o Supplicant is the Workstation or Client must be using 802.1x Client software.
- o Supplicant could be an end-user device, a printer, or an IP phone.
- o The supplicant is the 802.1x software that runs on the endpoint.
- o Windows has own common native supplicant for wired networks.

Authentication Server:

- o Authentication Server is a device that processes authentication such as RADIUS.
- o Authentication Server is the device that authenticates the Supplicant or client.
- o The entity that validates the identity of the supplicant and notifies the authenticator.
- o Authentication Server notifies authenticator to allow or deny the client request.
- o For example, RADIUS server, such as ACS, can provide authentication server services.

Authenticator (Switch, AP):

- o Device between supplicant & authentication server that facilitates authentication.
- o The client or supplicant is normally directly connected to the authenticator.
- o For example, switch or wireless access point provide authenticator services to clients.
- o Authenticator is the network device that is acting as a “gatekeeper” to the network.
- o Authenticator is typically a switch or Wireless LAN Controller (WLC).



802.1X Port States:

- o When using 802.1x, there are three defined port states:
- o Port state determines whether client will get the network access.

Auto:

- o Enables 802.1X port-based authentication & causes the port to begin in unauthorized state.
- o Auto port state allowing only EAPOL frames, CDP, and STP traffic to be sent and received.
- o After the supplicant is authenticated, the port transitions to the authorized state.

Forced-Authorized:

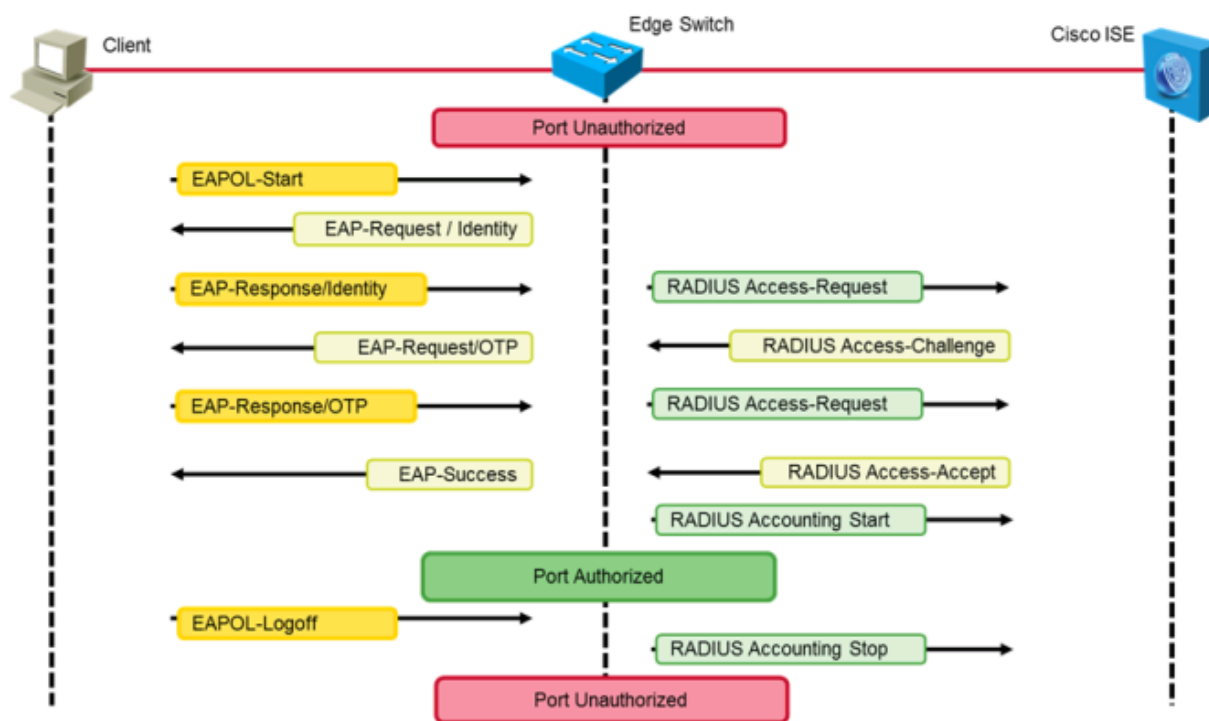
- o Forced Authorized state disables 802.1x port-based authentication.
- o Disables 802.1x and causes the port to transition to the authorized state.
- o In this state, all traffic is allowed as normal without restriction.
- o The port transmits & receives normal traffic without 802.1x-based authentication.

Forced-Unauthorized:

- o In this state, the port ignores all traffic, including any attempts to authenticate.
- o The port is forced to never authorize any connected client.
- o Port remain in the unauthorized state, ignoring all attempts by client to authenticate.
- o The authenticator cannot provide authentication services to the supplicants.

802.1X Port State	
802.1X Port State	Description
Force-Authorized	Client is always authorized to send traffic (Default)
Force-Unauthorized	Client is never authorized to send traffic (Even after authentication)
Auto	802.1x decides whether client is authorized or not to send traffic

Host Modes	Description	MAC
Single-host	Only one client can connected to the 802.1X enabled port	1
Multi-host	Allow multiple hosts useful for Access Point	*
Multi-domain	Allow both a host & VOIP Phone to be connected	2
Multi-auth	Allow one client Voice & multiple client data	*



802.1X Phasing:

- o There are three mode of IEEE 802.1X mention below.

Monitor Mode:

- o Monitor mode is the Phase 1 of 802.1x phasing.
- o Monitor Mode works like an audit mode.
- o Even failed authentication will allow access to network.
- o Administrator uses this mode to verify that all devices are authenticating.
- o Administrator uses this mode by using Logging data for verification.
- o Administrator get info, which users are, getting successful or failure authentications.
- o Failure authentications can be solved without affecting end user access to the network.
- o Authentication may be 802.1x or MAC Authentication Bypass (MAB).
- o Monitor mode uses RADIUS accounting packets and Open Authentication.
- o Monitor mode also uses RADIUS Multi Authentication feature to provide visibility.
- o Monitor Mode is address any possible authentication issues to moving to next phases.
- o Monitor Mode is that it is applicable to wired environments only.

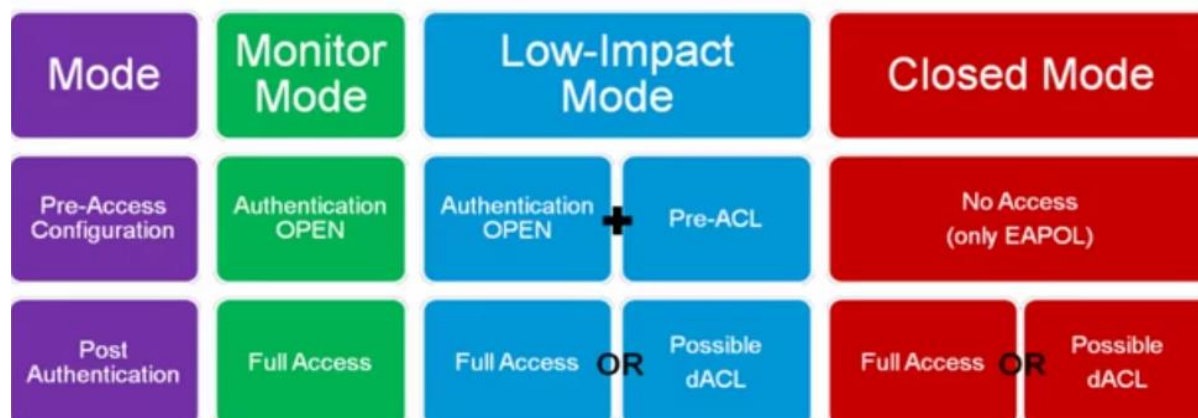
Low Impact Mode:

- o Low Impact Mode, Security is added over the framework built in Monitor mode.
- o Low Impact mode is also same as monitor mode with prebuilt ACL on switch port.
- o Low Impact Mode, Limited, basic access prior to authentication.
- o ACL restricts the port to very limited network access prior the authentication.
- o When user is authenticated successfully, additional resources may have granted.
- o Low Impact Mode, grant specific access after successful authentication.
- o In Low impact mode, host connected to the port may be allowed to use DHCP & DNS.
- o Low impact mode to route to the Internet and blocked to use internal resources.
- o In Low impact mode after authentication, a downloadable ACL may allow all traffic.

Closed Mode:

- o Closed mode is also lies in the 2nd phase.
- o Closed mode is formerly called High Security mode.
- o Closed mode, only EAPOL traffic is allowed before authentication.
- o Closed Mode, specific access after successful authentication.
- o Closed Mode is default 802.1X behavior & most restrictive method.
- o Any traffic before authentication will be dropped including DHCP, DNS, ARP etc.

IEEE 802.1X Wired Modes



IEEE 802.1X Wired Modes
Open Mode SW(config)#interface e0/0 SW(config-if)#authentication host-mode multi-auth SW(config-if)# authentication open SW(config-if)#authentication port-control auto SW(config-if)#mab SW(config-if)#dot1x pae authenticator
Low Impact Mode SW(config)#interface e0/0 SW(config-if)#authentication host-mode multi-auth SW(config-if)#authentication open SW(config-if)#authentication port-control auto SW(config-if)#mab SW(config-if)#dot1x ape authenticator SW(config-if)# ip access-group default-ACL in
Closed Mode SW(config)#interface e0/0 SW(config-if)#authentication host-mode multi-auth SW(config-if)# authentication port-control auto SW(config-if)#mab SW(config-if)#dot1x pae authenticator

Monitor Mode Authentication Open	Low Impact Mode Authentication Open + ACL
Closed Mode Remove Authentication Open and ACL	